

Plasmid expression of lasR from PA14 in pBBR1MCS2

Primer design • PCR / Assembly protocol • 2026-05-06

1 Konstrukt-Überblick

Zielgen	lasR in PA14 (720 nt, 239 aa)
Strategie	Plasmid expression (coding sequence 720 nt)
Vektor	pBBR1MCS2 (5148 bp), HindIII-linearisiert
Methode	In-Fusion 2-Fragment (Vektor + Insert)
Counterselektion	Km-Selektion (pBBR1MCS-2)
Finales Plasmid	pBBR1MCS2-lasR_PA14 • 5886 bp

2 Primer

Primer	Sequenz 5'→ 3' (Tail fett, Body normal)	nt	Tm _{body}	GC
P1_lasR_INSERT_Fwd	CGGTATCGATAAGCTAGGAGGACTTGTTTCATGGCCTTGTTGACGGTTTTTC	51	60.7 °C	50 %
P2_lasR_INSERT_Rev	ATTCGATATCAAGCTTCAGAGAGTAATAAGACCCAAATTAACG	43	57.0 °C	36 %

3 PCR-Bedingungen (B7 High Fidelity)

Tm _{body} Mittel	58.9 °C (Spread 3.72 °C)
Annealing Ta	59 °C (= Tm _{body} ; Gradient ± 5 °C empfohlen falls 1. Versuch fehlschlägt)
Elongation	60 s/kb • Insert 749 bp → 45 s

4 Konstrukt-Zusammenfassung

Insert (PCR-Amplikon) 749 bp • Final 5886 bp

Off-Target Scan: PASS • HindIII-Sites in finalem Plasmid: 1 (erwartet 1)